Agenda

Introduction
Agent Architectures
Platform Reference Architecture
Tools & Frameworks
Working Examples
Communication Protocols (MCP, A2A, ACP)
Best Practices

Why Agentic AI?

- LLMs evolving from passive tools to active agents
- Autonomous task orchestration
- Need for memory, reasoning, planning

AI Agents vs Agentic AI

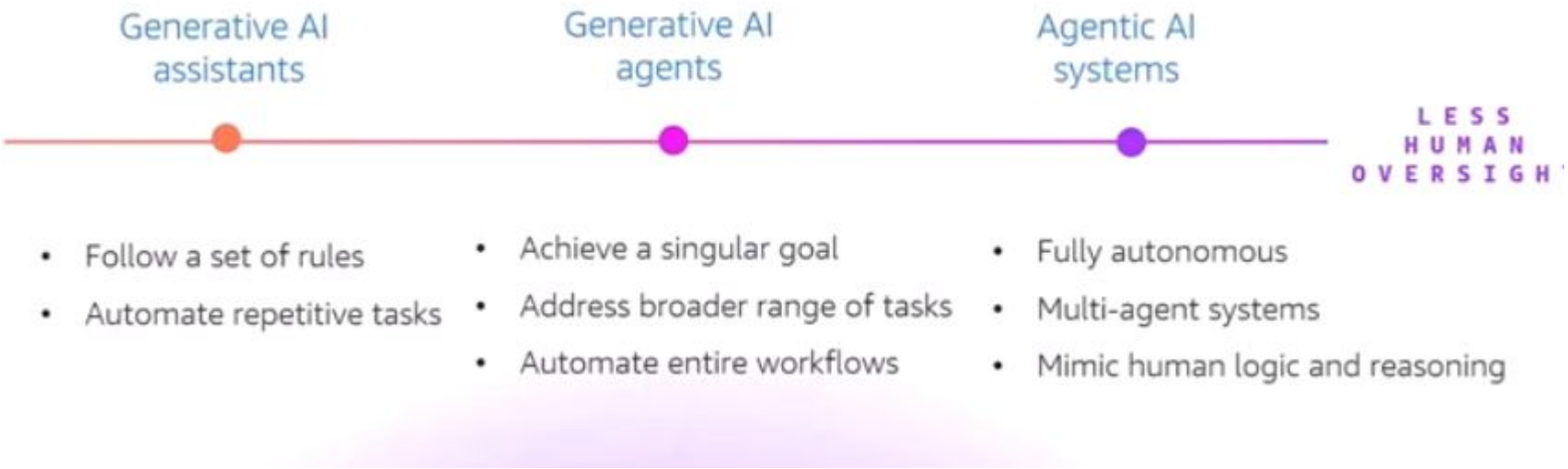
An **AI Agent** is an individual intelligent system that:

- **Perceives** → **Decides** → **Acts** to achieve a goal.
- It usually works **alone** and focuses on completing specific tasks.

Agentic AI is a **system-level paradigm** where:

- Multiple intelligent agents collaborate, plan, learn, and self-improve to achieve complex goals.
- It is about building ecosystems of agents, not just one.

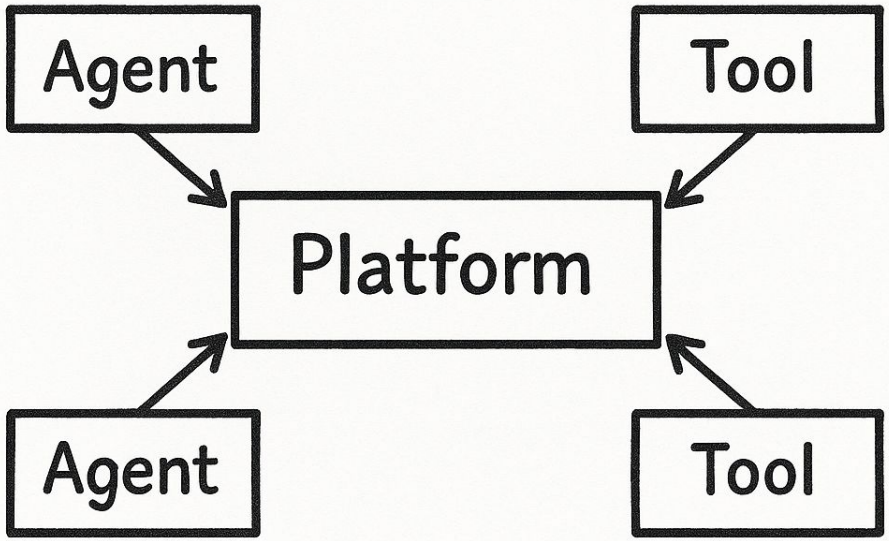
Evolution of Agentic AI



Reference Platform Architecture

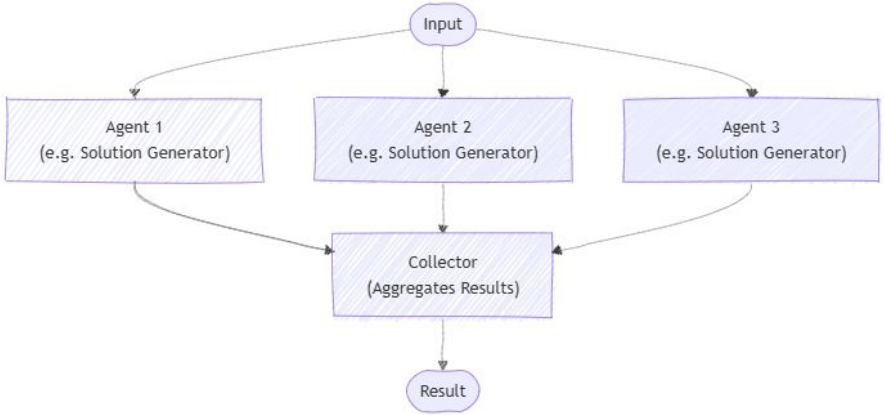
- **Planner:** Decomposes goals into actions or chains-of-thought. Can be rule-based or LLM-based (chain-of-thought prompts). Guides overall strategy
- **Memory:** Often split into *working memory* (short-term conversation context) and *persistent memory* (long-term vector store)
- **Execution/Tools:** Agents execute plans by calling external tools/APIs (e.g., search, databases, web APIs)
- **Observability:** Integrate logging, tracing, and metrics (e.g. via OpenTelemetry) to track agent decisions and performance
- **Environment:** The “perception” layer – how agents receive input (user requests, sensor data, events) and output results (UI, API responses).

Platform Architecture



Agent Patterns

- **Concurrent Pattern:** Multiple agents work on the same input in parallel.
- **Group-chat Pattern:** Agents and optional humans interact via chat interface.
- **Handoff Pattern:** Agents dynamically pass the control
- **Magentic Pattern:** A central manager selects which special agent to invoke next
- **Collector Pattern**

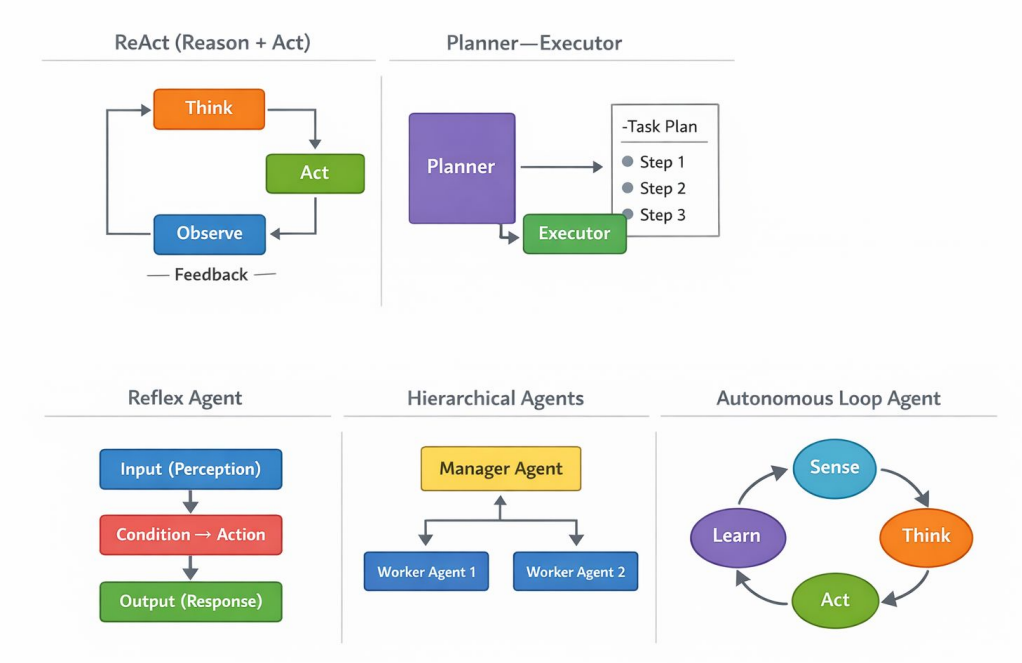


Agent patterns contd.,

Planner–Executor: Planner creates a plan and executor performs steps. A type of Hand-off pattern.

Reflex Agents: Respond to stimuli, they dont have memory. Used in chatbots, simple automation.

Hierarchical Agents: High-level agent delegates tasks. This is also a type of magentic pattern.

Autonomous Loop Agents: They have Continuous sense–think–act cycle

Agent Patterns